

TAXES AND SUBSIDIES PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

In this exercise you will redo some of the calculations that I did in class regarding **per-unit taxes**. The purpose of this exercise is for you *to practice some important algebraic manipulations regarding taxes*. Recall that when a tax is introduced into a market by the government, we must distinguish between the price paid by consumers, P^d , and the price received by sellers, P^s .

Suppose that the demand and supply equations are given by

$$Q = a - bP^d \quad (1)$$

$$Q = c + dP^s \quad (2)$$

where A, a and b are positive numbers and B can be any real number.

Let the per-unit tax be given by T . Then we know that

$$T = P^d - P^s \quad (3)$$

Using equations (1), (2) and (3), find the algebraic expressions for:

- (a) Equilibrium price without tax (the so called efficient market price). Denote this price by P^e
- (b) Equilibrium quantity without tax (the efficient market quantity). Denote this quantity by Q^e
- (c) The price paid by consumers after a positive tax $T > 0$ is imposed.
- (d) The price received by sellers after a positive tax $T > 0$ is imposed.
- (e) The after-tax equilibrium quantity. Denote this quantity by Q^T
- (f) The government revenue from the tax.
- (g) The tax incidence, that is calculate the ratios $(P^d - P^e)/T$ and $(P^e - P^s)/T$.
- (h) The dead-weight loss generated by the tax.

Problem 2

Now consider the case of a **per-unit subsidy**, $S > 0$. In this case we will have

$$S = P^s - P^d. \quad (4)$$

Repeat the same calculations as in part (a), that is, find the algebraic expressions for:

- (a) The price paid by consumers after the positive subsidy is imposed.

- (b) The price received by sellers after the positive subsidy is imposed.
- (c) the after-subsidy equilibrium quantity. Denote this quantity by Q^S .
- (d) The amount spent by the government to fund the tax.
- (e) The subsidy incidence, that is calculate the ratios $(P^e - P^d)/S$ and $(P^s - P^e)/S$.
- (f) The dead-weight loss generated by the subsidy.

Problem 3

Consider the expressions that you obtained in problems 1 and 2 and observe them carefully. They depend on the *distortion generated by the tax/subsidy* which is the after tax/subsidy equilibrium quantity. What happens to the quantities you calculated when:

- (a) We have a perfectly inelastic an inelastic demand and a regularly sloped supply (elastic), that is, when $a = 0$ and $b > 0$.
- (b) We have a perfectly inelastic supply and a regularly sloped demand (elastic), that is, when $a > 0$ and $b = 0$.
- (c) Observe in particular what happens to the dead-weight loss in each of these extreme cases. Conclude that whenever we have one of the extreme cases of either a perfectly inelastic demand or a perfectly inelastic supply, *the dead-weight loss is zero*.
- (d) Use the previous item your previous calculations to conclude that when we have one of these extreme cases, the side of the market that is perfectly inelastic pays the whole amount of the tax/receives the whole amount of the subsidy.

Problem 4

Now we will analyze the case of perfectly elastic demand or perfectly elastic supply.

- (a) A perfectly elastic demand corresponds to the case in which any quantity is consumed at a specific price P^* . Think about a horizontal line at P^* when graphing this demand equation. Note that the efficient equilibrium price will be precisely P^* . When supply is regularly sloped but demand is perfectly elastic, the price buyers pay will be $P^d = P^*$. The price sellers will receive will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that suppliers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic.
- (b) A perfectly elastic supply corresponds to the case in which any quantity is supplied at a specific price P^* . Think about a horizontal line at P^* when graphing this supply equation. Once again, note that the efficient equilibrium price will be P^* . When demand is regularly sloped but supply is perfectly elastic, the price sellers receive will be $P^s = P^*$. The price buyers will pay will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that buyers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic.

Problem 5

Consider the following scenarios: (demand and supply equations with one tax and one subsidy):

- i. $Q = 100 - 2P^d$, $Q = 10 + P^s$, a tax of $T = 10$, a subsidy of $S = 10$.
- ii. $Q = 10 - P^d$, $Q = -\frac{2}{3} + \frac{P^s}{3}$, a tax of $T = 4$, a subsidy of $S = 3$.
- iii. $Q = 95,000 - 50P^d$, supply is fixed at $Q = 90,000$, a tax of $T = 50$, a subsidy of $S = 20$.
- iv. $Q = 600 - P^d$, $Q = P^s$, a tax of $T = 300$, a subsidy of $S = 400$.
- v. $Q = 28 - P^d$, supply is $P = 20$, a tax of $T = 1$, a subsidy of $S = 10$.

For each one of these cases, compute

- (a) Amount of tax paid/amount of subsidy received by consumers.
- (b) Amount of tax paid/amount of subsidy received by sellers.
- (c) Government revenue from the tax/Government expenditure from the subsidy.
- (d) Consumer surplus before and after the tax/subsidy.
- (e) Producer surplus before and after the tax/subsidy.
- (f) Dead-weight loss from the tax/subsidy.
- (g) Define the following variables:

CS = consumer surplus

PS = producer surplus

$SS = CS + PS$ = social surplus

R = Government revenue from the tax

E = Government expenditure from the subsidy

DWL = Dead-weight loss

Observe carefully the calculations you made in each of the cases in this exercise. Use them to conclude that the following identities hold:

$$\text{For a tax: } SS^{\text{before tax}} = SS^{\text{after tax}} + R + DWL$$

$$\text{For a subsidy: } SS^{\text{after subsidy}} + DWL = SS^{\text{before subsidy}} + E$$

Use these identities to conclude that:

- Whenever there is a tax, part of the original social surplus is lost, that is, neither society nor the government appropriates it.
- Whenever there is a subsidy, government expenditures are covering the change in surplus to society plus an extra amount that nobody appropriates and thus is lost.